

001a — Book Abstract of the Complete Collection

FFGFT / T0 Theory — All Documents in Overview

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The T0 theory (time-mass duality) derives all fundamental constants and particle masses from a single dimensionless parameter $\xi = 4/3 \times 10^{-4}$ arising from the 4D identification torus geometry T^4 . This document is the summary of the complete document collection (over 250 documents) and includes from this version a new section on the epistemological origin of the theory.

Epistemological Origin: Four Methodological Convictions

Before the first formula was written, four methodological convictions stood at the beginning — not hypotheses to be tested, but prerequisites recognised as necessary.

1. No Fundamental Contradiction Between QM and RT

The actual starting point was not a formula but an epistemic stance: quantum mechanics and relativity are both good — too good to be fundamentally irreconcilable. Two theories that describe reality with such precision cannot have a genuine contradiction. The apparent conflict is a description artefact, not a natural phenomenon.

This conviction forced the search for a common geometric structure — and led to $T \cdot m = 1$ and thereby to ξ . $\hbar$ and c are in this framework SI conversion factors, not ontological primitives. The Higgs formula $\xi = \lambda_{\hbar^2 v^2} / (16\pi^3 m_{\hbar^2})$ and $\alpha = e^2 \mu_0 c / (4\pi \hbar)$ were moments of insight — not derivation sources. The Koide formula, α , and lepton masses were verifications, not starting points.

2. Instantaneity Must Be a Description Artefact

Bell tests demonstrate quantum correlations with seemingly instantaneous action across arbitrary distances. But relativity has locality as its foundational structure. If both theories are good, then *instantaneity must be an artefact of the description* — regardless of how convincingly the Bell tests appear to demonstrate otherwise.

The consequence: Bell correlations are real, but they are geometrically structured on T^4 , not ontologically nonlocal. Entanglement is a topological

connection, not action at a distance. The experimental predictions remain identical to quantum mechanics.

Key sentence from Dok. 230: Entanglement as a topological connection — not as a nonlocal correlation — the experimental predictions are the same.

3. The Simple Torus Is a Simplification

T^4 as a flat identification torus is already a strong simplification of reality. Curvature mechanisms on a non-flat T^4 can produce local geometric proximity for infinitely long light paths. What appears in the flat projection as spatially separated points can in the curved geometry be genuinely adjacent.

What appears as instantaneity could therefore be geometric proximity on a curved T^4 — local action along a long path invisible in the flat projection. This is structurally related to the ER-EPR approach [Maldacena/Susskind 2013], but without its string-theoretic prerequisites.

4. The Thought Itself Is a T^4 Process

The sharpest form of self-referentiality: the founding thought of FFGFT is itself a material process within T^4 . It arose in a physical brain processing structures of reality.

Information comes first — but only because matter comes first, making information processing possible. There is no pure thinking before matter and no pure matter without information processing. FFGFT arose from within T^4 , by a T^4 process, about T^4 processes.

The theory that says “we always describe from within” was itself developed from within. This is why “why T^4 ?” remains explicitly open — structural honesty, not a gap.

The Core Principle: Everything from One Number

Central Theorem: All physical constants — G , $\hbar$, c , e , all particle masses — can be derived from $\xi = 4/3 \times 10^{-4}$. From ξ follows $\alpha \approx 1/137.036$, particle masses and coupling constants without additional free parameters.

Foundations of T_0 Theory

Time-Mass Duality

The fundamental relation: $m(x) = \hbar/(c^2 \cdot T(x))$. Absolute time measure T_0 , variable mass, intrinsic T -field.

Derivation of All Constants

Gravitational constant G , fine structure constant α , lepton masses, Hubble constant $H_0 \approx 66.2$ km/s/Mpc — all derived from ξ , no further free parameter.

Cosmological Consequences

Static universe without expansion, without dark matter and without dark energy. CMB temperature, redshift and Hubble tension explained by T-field geometry.

Falsifiable Predictions

$L_0 = \xi \cdot \ell_P \approx 2.2 \times 10^{-39} \text{ m}$ | $D_f = 2.999867$ (gamma-ray burst dispersion) | No singularity in black holes | $\alpha^{-1} = 137.036$ (deviation $< 0.001 \%$) | a_e deviation reduced to 0.014% .

Document Structure

The collection comprises over 250 documents: Foundations, QM, QFT, Cosmology, Particle Physics, Experimental Tests, Technical Applications. GitHub: github.com/jpascher/T0-Time-Mass-Duality | Zenodo: [10.5281/zenodo.20117635](https://doi.org/10.5281/zenodo.20117635)

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